EE426 / 626 Microprocessor Applications

Examples of Embedded Firmware Errors Relating to Digital Data Overflow



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Hong Kong MTR Rail System

Tuesday, October 16, 2018

MTR Corporation – 4 Line Breakdown

A synchronization counter had been running for 20 years. Reached saturation and halted.

8 sector computers halted as designed. All of the 25 sector computers experienced instability.

Complete system needed to be reinitialized.



Hong Kong Central MTR Station, October 16
Fair use, South China Morning Post, Dec 19, 2018

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GPS 1024 Week Number Rollover

August 21-22, 1999
April 6-7, 2019
November 20-21, 2038

Legacy NAV (L1) Format:
- 10-bit week counter rolls from 1023 → 0
- (1024 weeks or 19.6 years)

Honda, Acura, Porsche 2022 → 2002

CNAV (L1C, L2C, L5)
- 13-bit counter rolls from 8191 → 0
- (8192 weeks or 157 years)



By U.S. Air Force - Original source: http://www.losangeles.af.mil/shared/media/ggallery/hires/AFG-100825-006.jpg
See also: https://www.af.mil/News/Photos.aspx?igphoto=2001638810
Full resolution: https://www.gps.gov/multimedia/images/GPS-III-A.jpg,
Public Domain, https://commons.wikimedia.org/w/index.php?curid=20220437

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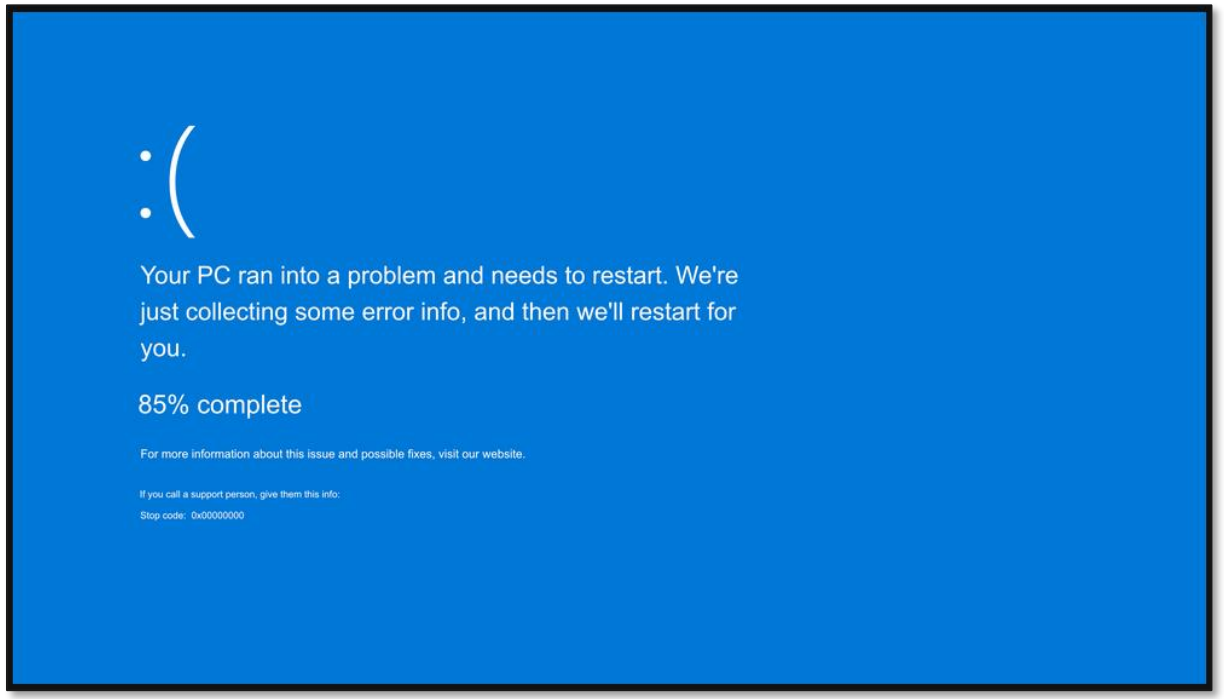
Windows 95 – 49.7 days

Virtual Device Driver (VxD)

32-bit counter (time in milliseconds)

Maximum time is 49.7 days

System will freeze without a reboot



Windows “Blue Screen of Death”

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Boeing 787 Counter Rollover

Electrical Power Control Units

Shutdown if powered without interruption for
248 days (approx. 8 months)

32-bit signed integer keeping 0.01 second intervals

248 days × 24 hrs/day × 60 min/hr × 60 s/min = 21,427,200
21,427,200 x 100 = 2,142,720,000

0x7FFFFFFF = Largest signed positive 32-bit integer
(2,147,483,647 approx. 248 days)



Fair use, Boeing 787 Dreamliner

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Intel Pentium FDIV Bug

October 24, 1994

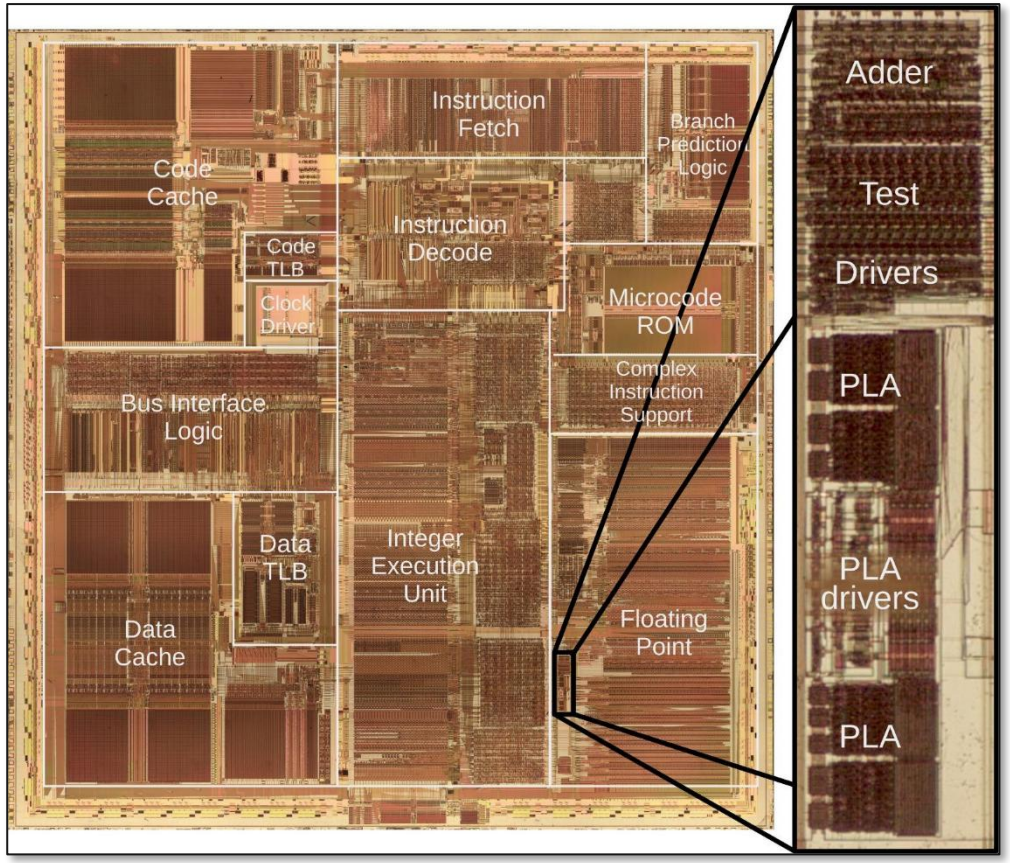
Mathematician Thomas Nicely discovers an FPU bug

$$\frac{4,195,835}{3,145,727} = 1.3338204491362410 02$$

$$\frac{4,195,835}{3,145,727} = 1.333739068902037589$$

Initially, Intel buries it, then calls it "*unimportant*"

Eventually Intel recalls ALL affected CPUs, **$475M**



Intel Pentium
Fair use, Image Credit Ken Shirriff